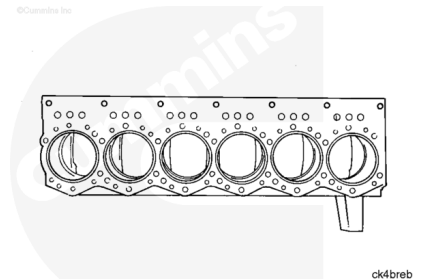


Trainings ▾

Clean

WARNING

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

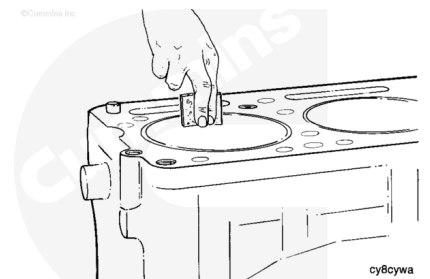


Clean the counterbore area. Use safety solvent.

Clean the top deck surface of the block. Use a scraper or a fibrous abrasive pad such as Scotch-Brite™ 7477, or equivalent, and solvent. Do **not** allow any dirt to get into the cylinder. The surface **must** be free of dirt, oil, and gasket material, but it does **not** have to look like new metal.

Check the top surface of the block for wear. If fretting damage is present in an area where a head gasket seal ring or a grommet makes contact, the surface **must** be repaired.

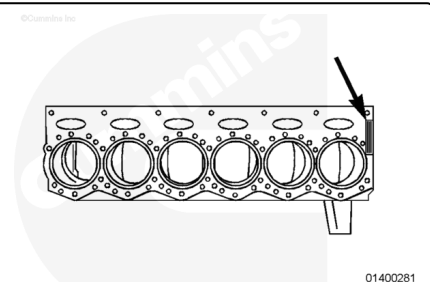
Fretting damage in any other area is acceptable **only** if it does **not** change the protrusion measurement of the counterbore or liner.



Inspect for Reuse

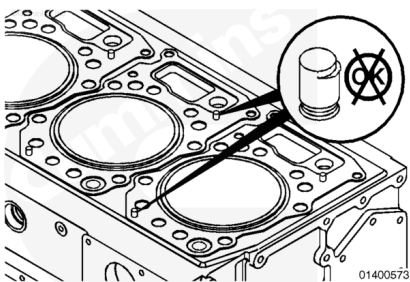
Inspect the cylinder block to determine if an oversize or standard size head gasket is required.

An oversize head gasket can be required on cylinder blocks that have had material removed from the top surface of the cylinder block. Check the cylinder block for markings indicating an oversize gasket is required.

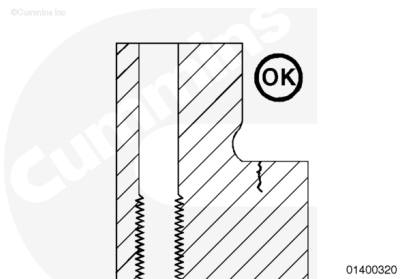


Inspect the groove pins for damage.

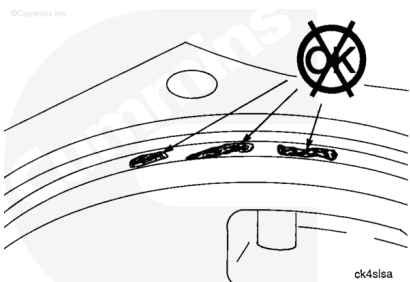
Replace damaged pins. Use dowel pin extractor, Part Number 3163720.



Metallurgical analysis of cross sections of counterbores having circumferential cracks has revealed the cracks are surface initiated on the top of the counterbore ledge, and normally do **not** propagate vertically through the counterbore ledge into the coolant passage around the cylinder liner.



Pitting on the liner seat is **not** acceptable. The graphic illustrates an example of pitting on the liner seat. Contact a Cummins® Authorized Repair Location if the block requires counterbore machining before it can be reused.

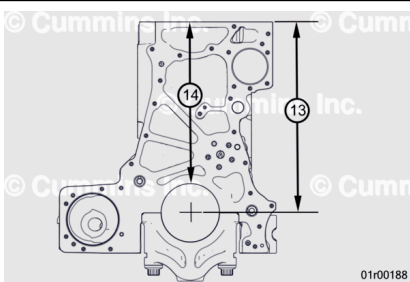


Measure

Inspect the cylinder block to determine which head gasket size is required.

The part line of the main bearing cap is **not** the same as the center line for the main bearing bore.

Measure the cylinder block height.



Standard Block Height			
	mm		in
From Crankshaft Center Line (13)	488.96	MIN	19.2503
	489.04	MAX	19.2535

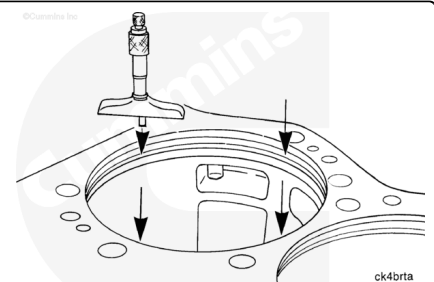
From Crankshaft Bore (14)	414.9475	MIN	16.3365
	415.0525	MAX	16.3406
Block Deck Height from Crankshaft Center Line (13) for +0.4 mm [0.016 in] Oversize Gasket	488.56	MIN	19.2346
	488.64	MAX	19.2377
Block Height From Crankshaft Bore (14) for +0.4 mm [0.016 in] Oversize Gasket	414.5475	MIN	16.3208
	414.6525	MAX	16.3249

The height of the cylinder block **must not** vary more than 0.08 mm [0.003 in] from end to end of the cylinder block. If the cylinder block height is **not** with specifications, the top surface of the cylinder block **must** be machined or the block must be replaced.

If the head deck surface of the cylinder block is machined, the ledge depth of the cylinder liner counterbore **must not** vary more than 0.025 mm [0.001 in] between the banks.

Make sure the micrometer contacts the flat surface of the ledge. It **must not** touch the radius.

Use a depth micrometer to measure the counterbore depth at the four locations illustrated in the graphic.



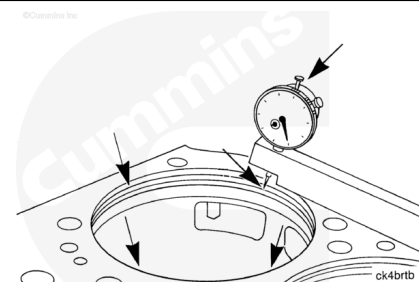
Counterbore Depth		
mm		in
14.00	MIN	0.551
14.05	MAX	0.553

The four measurements **must not** vary by more than 0.025 mm [0.001 in]. If the measurements exceed specifications, the counterbore **must** be machined. Contact a Cummins® Authorized Repair Location.

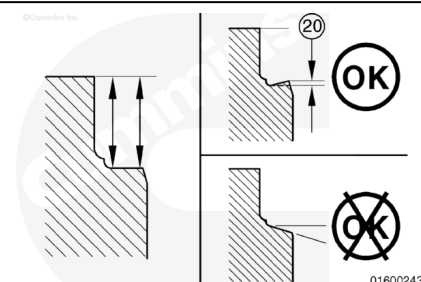
Make sure the indicator does **not** contact the counterbore radius on a block that does **not** have a double undercut.

Use depth gauge assembly, Part Number 3164438, to measure the angle of the counterbore ledge at four locations on the counterbore circumference.

The measurement of the ledge depth **must** be as near to the counterbore radius as possible, and as near to the counterbore edge as possible.



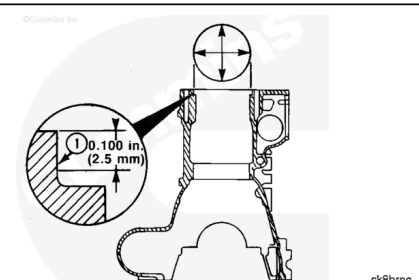
The angle (see callout number 20 in the illustration) of the counterbore ledge is acceptable if the measurement near the counterbore edge is the same or **not** more than 0.036 mm [0.001 in] shorter than the measurement near the counterbore radius. If the measurement near the counterbore ledge is longer than the measurement near the counterbore radius, the ledge **must** be machined.



Measure the inside diameter of the upper counterbore (1). The point of measurement **must** be within 2.54 mm [0.100 in] from the top of the block.

Counterbore Diameter - Upper Press Fit Diameter		
mm		in
205.93	MIN	8.107
205.99	MAX	8.110

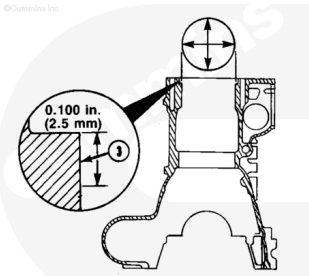
The inside diameter of the upper counterbore **must** be completely round within 0.050 mm [0.002 in]. If the measurement is **not** within specifications, check to see if the block can be machined to use oversize liners.



The upper counterbore **must not** be more than 0.076 mm [0.003 in] larger than the cylinder liner flange.

If the counterbore does **not** meet specifications, an oversize liner has possibly been used in a previous machining bore process. Measure and record the measurement so the proper press fit liner can be determined.

Measure the inside diameter of the lower counterbore (3) . The point of measurement **must** be within 2.54 mm [0.100 in] from the top of the counterbore ledge.



ck4brna

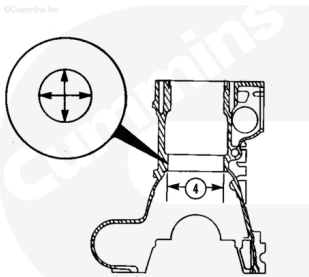
Counterbore Diameter - Lower Press Fit Diameter		
mm		in
194.48	MIN	7.657
194.54	MAX	7.659

The inside diameter of the lower counterbore **must** be completely round within 0.050 mm [0.002 in].

If the counterbore does **not** meet specifications, an oversize liner has possibly been used in a previous machining bore process. Measure and record the measurement so the proper press fit liner can be determined.

If the block is **not** within specifications, do **not** use the block.

Check the chamfer at the top of the packing ring bore (4) . Excessive pitting **must** be repaired.

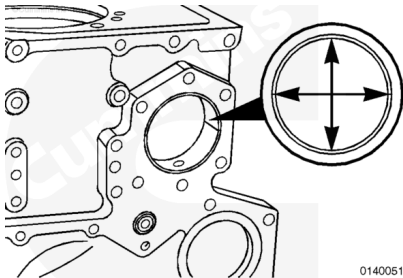


ck4brmb

Packing Ring Bore		
mm		in
190.34	MIN	7.494
190.40	MAX	7.496

Measure the inside diameter.

Camshaft Bushing Inside Diameter		
mm		in
105.01	MIN	4.134
105.10	MAX	4.138

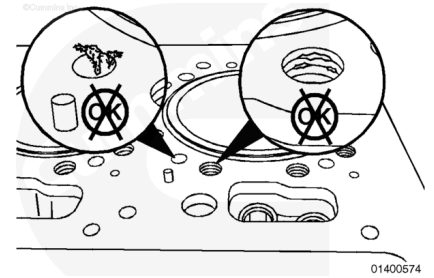


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If the bushing is **not** within specifications, the bushing **must** be replaced. Refer to Procedure 001-010 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-010.html>).

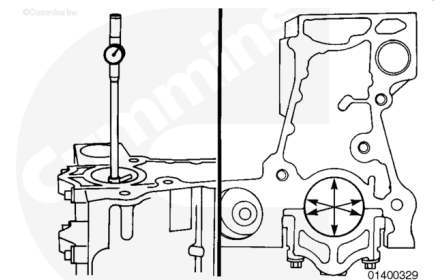
Check the coolant holes. If erosion or pitting is more than 0.08 mm [0.003 in] deep, or extends more than 2.41 mm [0.095 in] from the edge of the hole, the coolant hole **must** be repaired.

Check the threads of the capscrew holes for damage. Use a thread insert if a damaged capscrew hole **must** be repaired. Contact an Cummins® Authorized Repair Location for proper thread inserts.



Install the main bearing caps. Refer to Procedure 001-006 in Section 1. (</qs3/pubsys2/xml/en/procedures/89/89-001-006.html>)

Use a dial bore indicator to measure the inside diameter of each bearing bore in three positions, as shown in the illustration.



Main Bearing Bore Diameter

mm		in
147.999	MIN	5.8267
148-025	MAX	5.8277

If the main bearing bore is **not** within specification, the block **must not** be reused.